

CCNP Lab: IS-IS Integrated Services to Integrated Services Lab



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Purpose:

The objective of this lab was to design a network architecture consisting of three separate IS-IS areas—

Area 1, Area 2, and Area 3—each containing two routers. This setup was intended to help us implement and understand Multi-area IS-IS routing, with an emphasis on configuring and distinguishing between Level 1 and Level 1-2 routers to ensure effective inter-area communication.

In addition to building the network, the lab aimed to provide a foundational understanding of IS-IS in a point-to-point environment. We explored how IS-IS behaves differently from other link-state protocols like OSPF, especially when operating over IPv4. A key part of the lab involved examining how IS-IS establishes neighbor relationships using hello packets, along with practicing the essential commands used to verify these adjacencies.

Another important goal was to introduce us to more advanced IS-IS topics such as DIS (Designated Intermediate System) elections and the mechanics behind multi-area IS-IS design. Ultimately, we were expected to understand the protocol well enough to confidently describe the full configuration process—from initial setup to full deployment—as if explaining it to an expert, such as Radia Perlman, who originally influenced this protocol's development.

Background on Lab Concepts:

Intermediate System to Intermediate System (IS-IS) is an Interior Gateway Protocol (IGP) that operates as a link-state routing protocol. It is widely recognized for its scalability and fast convergence. Like OSPF, IS-IS uses Dijkstra's algorithm to determine the shortest path between nodes in a network.

IS-IS routers can operate in three modes:

- Level 1: Communicates only within its own area.
- Level 2: Communicates only between different areas.
- Level 1-2: Can communicate both within its own area and across other areas, making it essential for connecting multiple IS-IS areas.

Although IS-IS and OSPF are both link-state IGPs that use similar concepts such as link-state databases, SPF (Shortest Path First), aging mechanisms, and metric-based routing decisions, there are key differences. IS-IS is generally more scalable and flexible in its area design. Unlike OSPF, which requires a strict backbone area (Area 0), IS-IS allows more adaptable area structures. Additionally, IS-IS operates directly over Layer 2 (the data link layer), adding a layer of inherent security and simplifying/reducing its IP dependency.

One notable operational difference is in their approach to router elections in broadcast networks: IS-IS elects a single DIS (Designated Intermediate System) which can be preempted, while OSPF elects a DR (Designated Router) and BDR (Backup Designated Router), which cannot be preempted. Our lab did not include DIS functionality since we focused on point-to-point links rather than multi-access networks, where DIS would typically apply.

IS-IS does not support NBMA (Non-Broadcast Multi-Access), point-to-multipoint, or virtual links, in contrast to OSPF's more extensive support for those topologies.

Each IS-IS router is identified using a NET (Network Entity Title), which consists of three components:

- Area ID: Typically 6 digits (e.g., 49.0002)
- System ID: A unique 12-digit identifier (e.g., 0000.0000.0004)
- N-Selector: Always set to 00 for adjacency formation

A full NET ID would something like: 49.0002.0000.0000.0004.00

Finally, IS-IS uses Dijkstra's algorithm to calculate optimal routes through the network. In essence, this algorithm repeatedly selects the nearest unvisited node and updates the shortest known paths to neighboring nodes until all nodes have been visited and the shortest paths from the source to all others are known.

Lab Summary:

What was set up:

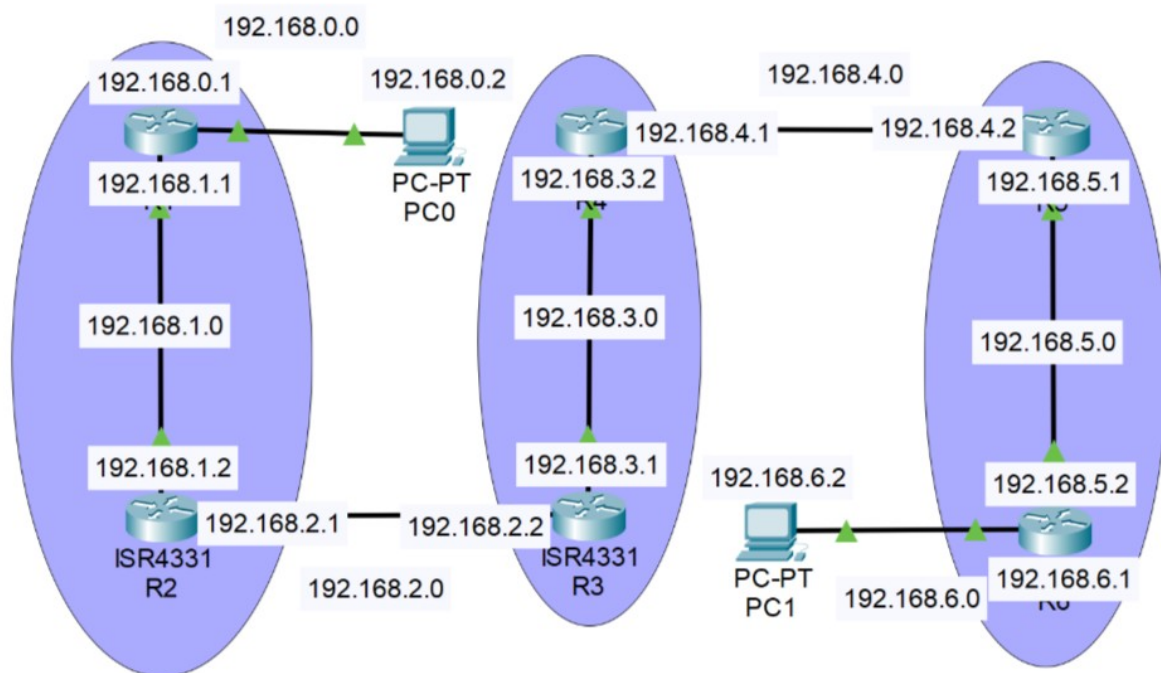
R1 (Level 1 IS-IS router)
R2 (Level 1-2 IS-IS router)
R3 (Level 1-2 IS-IS router)
R4 (Level 1-2 IS-IS router)
R5 (Level 1-2 IS-IS router)
R6 (Level 1 IS-IS router)

PC1 connected to R6

R1 connected to R2 (area 1)
R2 connected to R3 (area 1 to area 2)
R3 connected to R4 (area 2)
R4 connected to R5 (area 2 to area 3)
R5 connected to R6 (area 3)

R6 connected PC2

Network Diagram:



Lab Commands:

show ip route – Displays the IPv4 routing table

ip routing – A command to enable the sending of data packets for IPv4 across an IP network

traceroute [destination ip address] - Figuring the data hops from the source ip address to the destination ip address.

ip router isis – Enables IS-IS as an IP routing protocol and enters router configuration mode for the protocol

show ip protocol - displays information about the parameters and settings of active IPv4 routing protocols running on the router

show isis neighbors – displays the status of all IS-IS neighbors configured on the router
Is-type

net [Area-ID. System-ID. N-Selector] Example Net-id: 49.0002.0000.0000.0004.00 (49.0002 is the Area-ID, 0000.0000.0004 is the system-ID, 00 is the N-Selector) - The Net-id acts a unique identifier for the router within the IS-IS domain

is-type level- [level number]- sets the routing level for an IS-IS instance.

is-type level-1 – configures a router that operates within an area that runs IS-IS

is-type level-2 – configures a router to act solely as a router that communicates between areas, participating in only level 2 routing with IS-IS

is-type level-1-2 – configures a router to act both as a router that operates within an area and between routers with IS-IS

metric-style narrow – the older, default metric type used for calculating costs between routers.

show isis protocol – displays information about IS-IS protocol including its configuration and status.

Configurations:

R1 configuration:

```
interface GigabitEthernet0/0/0
 ip address 10.0.0.4 255.255.255.0
 ip router isis
 negotiation auto

interface GigabitEthernet0/0/1
 ip address 192.168.0.1 255.255.255.0
 ip router isis
 negotiation auto

interface GigabitEthernet0/2/0
 no ip address
 negotiation auto

interface GigabitEthernet0/2/1
 no ip address
 negotiation auto

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto

router isis
 net 49.0001.0000.0000.0001.00
 is-type level-1
 metric-style narrow
```

R2 Configuration:

```
interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.0
```

```
ip router isis
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
ip router isis
negotiation auto

interface GigabitEthernet0/2/0
no ip address
negotiation auto

interface GigabitEthernet0/2/1
no ip address
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto

router isis
net 49.0001.0000.0000.0002.00
metric-style narrow

ip forward-protocol nd
no ip http server
ip http secure-server
```

R3 Configuration:

```
interface GigabitEthernet0/0/0
ip address 192.168.3.1 255.255.255.0
ip router isis
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.0
ip router isis
negotiation auto

interface Serial0/1/0
no ip address
shutdown

interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

interface Vlan1
 no ip address
 shutdown

router isis
 net 49.0002.0000.0000.0003.00
 ip forward-protocol nd
 no ip http server
 no ip http secure-server
 ip tftp source-interface GigabitEthernet0
```

R4 Configuration:

```
interface GigabitEthernet0/0/0
 ip address 192.168.3.2 255.255.255.0
 ip router isis
 negotiation auto
```

```
interface GigabitEthernet0/0/1
 ip address 192.168.4.1 255.255.255.0

 ip router isis
 negotiation auto
```

```
interface Serial0/1/0
 no ip address
```

```
interface Serial0/1/1
 no ip address
```

```
interface GigabitEthernet0

 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto
interface Vlan1 no ip address
```

```
router isis
 net 49.0002.0000.0000.0004.00
```

```
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
```

R5 Configuration:

```
interface GigabitEthernet0/0/0
ip address 192.168.5.1 255.255.255.0
```

```
ip router isis
negotiation auto
```

```
interface GigabitEthernet0/0/1
ip address 192.168.4.2 255.255.255.0
```

```
ip router isis
negotiation auto
```

```
interface Serial0/1/0
no ip address
```

```
interface Serial0/1/1
no ip address
```

```
interface GigabitEthernet0

vrf forwarding Mgmt-intf
no ip address
negotiation auto
```

```
interface Vlan1
```

```
no ip address
```

```
router isis
net 49.0003.0000.0000.0005.00
```

R6 Configuration:

```
interface GigabitEthernet0/0/0
ip address 192.168.5.2 255.255.255.0
```

```

ip router isis
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.6.1 255.255.255.0
ip router isis
negotiation auto

interface Serial0/1/0
no ip address
shutdown

interface Serial0/1/1
no ip address
shutdown

interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto

interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router isis
net 49.0003.0000.0000.0006.00
is-type level-1
metric-style narrow

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

```

Show IP Protocol for IPv4:

R1:

```
R1#show isis protocol
```

```

IS-IS Router: <Null Tag> (0x10000)
  System Id: 0000.0000.0001.00  IS-Type: level-1
  Manual area address(es):

```

```
49.0001
Routing for area address(es):
49.0001
Interfaces supported by IS-IS:
GigabitEthernet0/0/1 - IP
GigabitEthernet0/0/0 - IP
Redistribute:
static (on by default)
Distance for L2 CLNS routes: 110
RRR level: none
Generate narrow metrics: level-1-2
Accept narrow metrics: level-1-2
Generate wide metrics: none
Accept wide metrics: none
```

R2:

```
R2#show isis protocol
```

```
IS-IS Router: <Null Tag> (0x10000)
System Id: 0000.0000.0002.00 IS-Type: level-1-2
Manual area address(es):
49.0001
Routing for area address(es):
49.0001
Interfaces supported by IS-IS:
GigabitEthernet0/0/1 - IP
GigabitEthernet0/0/0 - IP
Redistribute:
static (on by default)
Distance for L2 CLNS routes: 110
RRR level: none
Generate narrow metrics: level-1-2
Accept narrow metrics: level-1-2
Generate wide metrics: none
Accept wide metrics: none
```

R3:

```
R3#show isis protocol
```

```
IS-IS Router: <Null Tag>
System Id: 0000.0000.0003.00 IS-Type: level-1-2
Manual area address(es):
49.0002
Routing for area address(es):
```

```
49.0002
Interfaces supported by IS-IS:
  GigabitEthernet0/0/1 - IP
  GigabitEthernet0/0/0 - IP
Redistribute:
  static (on by default)
Distance for L2 CLNS routes: 110
RRR level: none
Generate narrow metrics: level-1-2
Accept narrow metrics:   level-1-2
Generate wide metrics:   none
Accept wide metrics:     none
```

R4:

```
R4#show isis protocol
```

```
System Id: 0000.0000.0004.00  IS-Type: level-1-2
Manual area address(es):
  49.0002
Routing for area address(es):
  49.0002
Interfaces supported by IS-IS:
  GigabitEthernet0/0/1 - IP
  GigabitEthernet0/0/0 - IP
Redistribute:
  static (on by default)
Distance for L2 CLNS routes: 110
RRR level: none
Generate narrow metrics: level-1-2
Accept narrow metrics:   level-1-2
Generate wide metrics:   none
Accept wide metrics:     none
```

R5:

```
R5#show isis protocol
```

```
IS-IS Router: <Null Tag>
System Id: 0000.0000.0005.00  IS-Type: level-1-2
Manual area address(es):
  49.0003
Routing for area address(es):
  49.0003
Interfaces supported by IS-IS:
  GigabitEthernet0/0/1 - IP
  GigabitEthernet0/0/0 - IP
```

```
Redistribute:
  static (on by default)
Distance for L2 CLNS routes: 110
RRR level: none
Generate narrow metrics: level-1-2
Accept narrow metrics:   level-1-2
Generate wide metrics:   none
Accept wide metrics:     none
```

R6:

```
R6#show isis protocol
IS-IS Router: <Null Tag> (0x10000)
  System Id: 0000.0000.0006.00  IS-Type: level-1
  Manual area address(es):
    49.0003
  Routing for area address(es):
    49.0003
  Interfaces supported by IS-IS:
    GigabitEthernet0/0/1 - IP
    GigabitEthernet0/0/0 - IP
  Redistribute:
    static (on by default)
  Distance for L2 CLNS routes: 110
  RRR level: none
  Generate narrow metrics: level-1-2
  Accept narrow metrics:   level-1-2
  Generate wide metrics:   none
  Accept wide metrics:     none
```

Show IP route/Traceroute for IPv4:

R1:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
      ia - IS-IS inter area, * - candidate default, U - per-user
static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
```

a - application route
+ - replicated route, % - next hop override, p - overrides
from PfR

Gateway of last resort is not set

```
i L2 192.168.1.0/24 [115/20] via 192.168.2.1, 3d00h,
GigabitEthernet0/0/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L    192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
i L1 192.168.4.0/24 [115/20] via 192.168.3.2, 3d00h,
GigabitEthernet0/0/0
i L2 192.168.5.0/24 [115/30] via 192.168.3.2, 3d00h,
GigabitEthernet0/0/0
```

R2:

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route

+ - replicated route, % - next hop override, p - overrides

from PfR

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.1.2/32 is directly connected, GigabitEthernet0/0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L    192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
i L2 192.168.3.0/24 [115/20] via 192.168.2.2, 3d00h,
GigabitEthernet0/0/1
i L2 192.168.4.0/24 [115/30] via 192.168.2.2, 3d00h,
GigabitEthernet0/0/1
```

```
i L2 192.168.5.0/24 [115/40] via 192.168.2.2, 3d00h,  
GigabitEthernet0/0/1
```

R3:

```
R3#show ip route  
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -  
BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
        E1 - OSPF external type 1, E2 - OSPF external type 2  
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS  
level-2  
        ia - IS-IS inter area, * - candidate default, U - per-user  
static route  
        o - ODR, P - periodic downloaded static route, H - NHRP, l -  
LISP  
        a - application route  
        + - replicated route, % - next hop override, p - overrides  
from Pfr
```

Gateway of last resort is not set

```
i L2 192.168.1.0/24 [115/20] via 192.168.2.1, 3d00h,  
GigabitEthernet0/0/1  
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks  
C      192.168.2.0/24 is directly connected, GigabitEthernet0/0/1  
L      192.168.2.2/32 is directly connected, GigabitEthernet0/0/1  
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks  
C      192.168.3.0/24 is directly connected, GigabitEthernet0/0/0  
L      192.168.3.1/32 is directly connected, GigabitEthernet0/0/0  
i L1 192.168.4.0/24 [115/20] via 192.168.3.2, 3d00h,  
GigabitEthernet0/0/0  
i L2 192.168.5.0/24 [115/30] via 192.168.3.2, 3d00h,  
GigabitEthernet0/0/0
```

R4:

```
R4#show ip route  
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -  
BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
        E1 - OSPF external type 1, E2 - OSPF external type 2  
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS  
level-2  
        ia - IS-IS inter area, * - candidate default, U - per-user  
static route  
        o - ODR, P - periodic downloaded static route, H - NHRP, l -  
LISP  
        a - application route
```

+ - replicated route, % - next hop override, p - overrides
from PFR

Gateway of last resort is not set

```
i L2 192.168.1.0/24 [115/30] via 192.168.3.1, 3d00h,
GigabitEthernet0/0/0
i L1 192.168.2.0/24 [115/20] via 192.168.3.1, 3d00h,
GigabitEthernet0/0/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/0
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.4.1/32 is directly connected, GigabitEthernet0/0/1
i L2 192.168.5.0/24 [115/20] via 192.168.4.2, 4d00h,
GigabitEthernet0/0/1
```

R5:

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route

+ - replicated route, % - next hop override, p - overrides
from PFR

Gateway of last resort is not set

```
i L2 192.168.1.0/24 [115/40] via 192.168.4.1, 3d00h,
GigabitEthernet0/0/1
i L2 192.168.2.0/24 [115/30] via 192.168.4.1, 3d00h,
GigabitEthernet0/0/1
i L2 192.168.3.0/24 [115/20] via 192.168.4.1, 3d00h,
GigabitEthernet0/0/1
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.4.2/32 is directly connected, GigabitEthernet0/0/1
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/24 is directly connected, GigabitEthernet0/0/0
L 192.168.5.1/32 is directly connected, GigabitEthernet0/0/0
```

R6:

R6#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route

+ - replicated route, % - next hop override, p - overrides

from Pfr

Gateway of last resort is 192.168.5.1 to network 0.0.0.0

i*L1 [0.0.0.0/0](#) [115/10] via 192.168.5.1, 4d00h, GigabitEthernet0/0/0

i L1 [192.168.4.0/24](#) [115/20] via 192.168.5.1, 4d00h,

GigabitEthernet0/0/0

[192.168.5.0/24](#) is variably subnetted, 2 subnets, 2 masks

C [192.168.5.0/24](#) is directly connected, GigabitEthernet0/0/0

L [192.168.5.2/32](#) is directly connected, GigabitEthernet0/0/0

Traceroute:

R1 to R6

R1#traceroute 192.168.5.2

Type escape sequence to abort.

Tracing the route to 192.168.5.2

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.2 1 msec 1 msec 1 msec

2 192.168.2.2 0 msec 1 msec 1 msec

3 192.168.3.2 1 msec 1 msec 1 msec

4 192.168.4.2 0 msec 1 msec 1 msec

5 192.168.5.2 1 msec 2 msec *

R6 to R1

R6#traceroute 192.168.1.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.5.1 1 msec 1 msec 0 msec

```

2 192.168.4.1 1 msec 1 msec 0 msec
3 192.168.3.1 1 msec 1 msec 0 msec
4 192.168.2.1 1 msec 1 msec 1 msec
5 192.168.1.1 2 msec 1 msec *

```

show isis neighbors:

R1:

```
R1#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R2 R1.01	L1	Gi0/0/0	192.168.1.2	UP	26

R2:

```
R2#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R1 R1.01	L1	Gi0/0/0	192.168.1.1	UP	7
R3 R2.02	L2	Gi0/0/1	192.168.2.2	UP	24

R3:

```
R3#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R2 R2.02	L2	Gi0/0/1	192.168.2.1	UP	9
R4 R3.01	L1	Gi0/0/0	192.168.3.2	UP	23
R4 R3.01	L2	Gi0/0/0	192.168.3.2	UP	23

R4:

```
R4#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R3 R3.01	L1	Gi0/0/0	192.168.3.1	UP	7
R3 R3.01	L2	Gi0/0/0	192.168.3.1	UP	9
R5 R5.02	L2	Gi0/0/1	192.168.4.2	UP	6

R5:

```
R5#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R4 R5.02	L2	Gi0/0/1	192.168.4.1	UP	22
R6 R5.01	L1	Gi0/0/0	192.168.5.2	UP	26

R6:

```
R6#show isis neighbors
```

System Id Circuit Id	Type	Interface	IP Address	State	Holdtime
R5 R5.01	L1	Gi0/0/0	192.168.5.1	UP	9

Problems:

One of the initial difficulties in the lab was configuring the IS-IS NET ID correctly. Unlike OSPF, which uses a much simpler Area ID, IS-IS requires a more complex format involving an NET, including an Area ID, a System ID, and an N-Selector. This unfamiliar structure led to some early connectivity issues until we were able to understand how the NET ID must be properly formatted in order for IS-IS adjacency to

form. Another challenge was seeing the difference between the different IS-IS router levels, Level 1, Level 2, and Level 1-2 and deciding which routers should be assigned which. Since the network consisted of three separate areas, selecting the correct router types was crucial for enabling inter-area communication across a point-to-point topology and prevented connections if improperly configured.

Concepts we learned:

This lab helped clarify the many differences between IS-IS and OSPF, many of which were discussed in the background section. We learned how to correctly construct and configure NET IDs for IS-IS routers and how critical that configuration is for the protocol to operate as intended.

We also gained a solid understanding of the roles of different IS-IS levels:

- **Level 1** routers operate strictly within a single IS-IS area.
- **Level 2** routers handle inter-area communication.
- **Level 1-2** routers function in both intra- and inter-area roles and are essential for connecting multiple areas together.

We learned that, unlike OSPF where each interface on a router can belong to a different area, IS-IS applies area assignments to the router as a whole—all interfaces on a router belong to the same area. This was an important conceptual shift that affected how we designed the topology. This helps with simplicity when designing networks, however could be proven to cause issues when creating more complex networks that require more flexibility in link connections.

What caused issues:

The biggest thing that caused issues in the beginning was the configuration of the is-type and thus failing to achieve connectivity between areas when we were only using is-type level 1 routers. We would eventually solve the problem by implementing level-1-2 routers between them. Another problem that caused issues was us forgetting the implementation of a Net-ID and also incorrect configuration of it at first, having no clue of how to configure it. After doing some research, we realize that the correct way to implement it is to set it with an Area-ID then System-ID then N-Selector. Also, we tried to set the different interfaces of a router to separate areas like in OSPF which ended up being the wrong way to configure for IS-IS, as IS-IS applies the area to every port on the router.

Conclusion:

In conclusion, through this lab where we had to create a network that has 3 areas that each have 2 routers in each separate area (area 1, area 2, and area 3) to route Multi-area IS-IS through the three areas. The concepts we're expected to learn include the different use of Level 1 and Level-1-2 routers to ensure the success of the IS-IS network as well as the NET IDs to identify each router and also to troubleshoot the network to ensure connectivity.

Teacher Signoff Page of Lab Complete:

IS-IS Integrated Services to Integrated Services Lab

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P3-4-8 CCNP Mr. Mason

